Weekly Report

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1 PPT on basics

I spoke to Ashutosh and in our discussion he told me to prepare a PowerPoint presentation covering the following topics:

1. Entanglement,
2. Generation using non-linear crystals,
3. Phase matching in BBO (Beta Barium Borate) crystals,
4. Quasi-phase matching in PPLN (Periodically Poled Lithium Niobate) crystals, and
5. Characterization of entanglement.
6. multiplexing of classical and quantum light,

I have learned the fundamentals of quantum entanglement. Entanglement implies that the state of one photon is always dependent on the other.

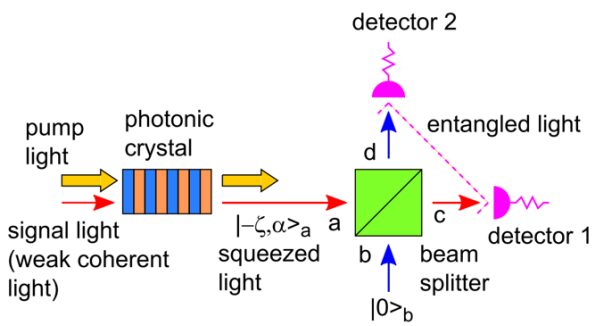

Figure 1: Generation of Entangled Photons

To generate entangled photon pairs, spontaneous parametric down-conversion (SPDC) is used, where a photon enters a nonlinear crystal and splits into two lower-energy photons (signal and idler), with the detection of one indicating the existence of the other. The use of one-dimensional photonic crystals with high second-order nonlinearity ($\chi^{(2)}$) and beam splitters enables efficient entangled photon generation, especially when using squeezed light, which reduces uncertainty in one of the beam from beam splitter.

Birefringent crystals like BBO allow phase matching through polarization and propagation direction tuning. Periodic poling in crystals (e.g., PPLN) supports quasi-phase matching by flipping nonlinear polarity at each coherence length to constructively interfere the waves. The effectiveness of SPDC is based on maximizing $\chi^{(2)}$ interaction and satisfying phase-matching.

To characterize entanglement, we use metrics like Schmidt rank, Von Neumann entropy (for pure states), and concurrence, negativity, logarithmic negativity, and entanglement of formation (for mixed states). Experimental methods like quantum state tomography and Bell inequality tests verify entanglement, while Hong-Ou-Mandel (HOM) interference demonstrates in distinguishability of photons.

In fiber optics, we learned that by combining weak quantum and strong classical signals (multiplexing) we find challenges like Raman scattering noise. Solutions include attenuation, large wavelength spacing (placing quantum in O-band and classical in C-band), and using Dense Wavelength Division Multiplexing (DWDM) filters. Coexistence of quantum and classical channels is important for scalable DV-QKD deployment but is limited by power budgets and interference.

Finally, I learned that the time evolution of a closed quantum system is governed by the Schrödinger equation, and the state transition is represented by a unitary operator dependent on time intervals, which can be assumed as known for single qubits.

After completing the presentation, I discussed it with Ashutosh to clarify few conceptual doubts. He gave feedback and, on the basis of his suggestions, I revised the presentation.

2 Characterization of Entanglement

In further discussions and after exploring different projects in lab, i opted to go for doing the characterization of entanglement experiment for my internship. Ashutosh has been guiding me and instructed to cover theory on

1. State vector formalism
2. Density matrix
3. Measurement
4. Beam splitter

To do this i followed the Nielsen and chuang book-chapter 2 , Where i learnt the postulates of Quantum mechanics.
Postulate 1 talks about how the state change can be represented as the multiple of the previous state and a unique Unitary matrix. the unitary matrices (for single qubits) are often chosen from the basis of the pauli matrices, after which its just simple matrix multiplication.
The second Postulate says how the evolution of states with change in time can be expressed with schrodingers equation.
The thrid postulate is about measurement. The general measurement method.
Quantum measurements are described by a set of measurement operators $\{M_m\}$, which act on the state space of the quantum system. The index m corresponds to the possible outcomes of the measurement.

If the system is in state $|\psi\rangle$ immediately before the measurement, the probability of obtaining outcome m is given by:

$$p(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$$

After obtaining result m, the state of the system collapses to:

$$|\psi'\rangle = \frac{M_m |\psi\rangle}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$$

The measurement operators satisfy the *completeness relation*:

$$\sum_m M_m^\dagger M_m = I$$

This completeness condition ensures that the total probability of all possible outcomes sums to one:

$$\sum_m p(m) = \sum_m \langle\psi| M_m^\dagger M_m |\psi\rangle = 1$$

Since this must hold for all $|\psi\rangle$, it is equivalent to the completeness equation above. In practice, verifying the completeness equation directly is more convenient.
Along with general measurement i covered the Projection and POVM measurement and solved few problems to test my understanding.
I have also learned the difference between classical and quantum beam splitters.
Due to lack of in depth background on quantum mechanics and no experience in quantum optics, i struggled to see the math with a physical meaning, i have taken help from Nilesh a little to try and build a physical intuition behind a few concepts. I have also addressed this issue to Ashutosh again and he has kindly shared his notes on beam splitters to help me. I am trying to get more familiar with the math by solving problems from the book "Problems and Solutions In Quantum Computing and Quantum Information - Willi-Hans Steeb and Yorik Hardy".

3 PSB characterization

I worked on characterizing the performance of Polarizing Beam Splitters (PBS).
Using a laser from AeroDIODE emitting $|H\rangle + |V\rangle$ at 5 dBm for 32dB input. I calculated power for all PSB's at their ports and compared them with the values that came when done previously at lab.

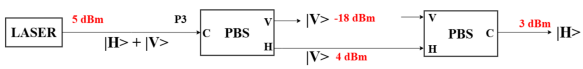

Figure 2: Characterization of PSB

I learned that when the light goes through P3 in PSB, the light will first rotate and then polarize but when it goes through P2 it will first polarise and then rotate.
Experimentally checked Validated whether the output is fast axis port, checked for blocking of fast axis and fed fast axis in slow axis port.
I also calculates polarization at every point for different set ups including the Faraday mirror, 50:50 beam splitter and phase modulator.

4 Weekly Plan

- Learn how to work with fiber based beam splitters
- Learn characterization of the same